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HCMC UNIVERSITY OF SCIENCE

FACULTY OF INFORMATION TECHNOLOGY



HOMWORK REPORT

COURSE: SOFTWARE TESTING

Assignment: Domain Testing on EShop

STUDENT INFORMATION

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1. DOMAIN TESTING

In the domain testing phase, I implemented the following AI techniques:

- The first step is to upload the README.md file (describing the Eshop system) and the S04_Domain Testing.pdf file (containing the content for Domain Testing) so that the AI can read and understand the contents before performing the required domain testing steps. The AI has provided a summary so I can check if it has grasped the general content of these two files.
- After the AI finishes reading the documentation, I instruct it to review the requirements for the FR section I selected for testing in the README.md file. It should determine the equivalent class for each condition stated in this file, and this determination must be closely aligned with the instructions in the S04_DomainTesting.pdf file. After the AI generates the classes, I review their output to see if they are appropriate and complete compared to the descriptions in the README. If not, I request that it adjust or add any missing elements that I have identified to its output to achieve the widest possible coverage.
- After the AI generated equivalence classes and I adjusted or added to them through my supplementary prompts, based on what had been done in the previous step, I continued to ask it to design test cases to cover the equivalence classes using the provided PDF document, specifically adhering to the following rules:
 - Choose at least one test case from each equivalence class.
 - For valid classes, choose test cases to cover as many equivalence classes as possible until all valid classes have been covered.
 - For invalid classes, choose test cases so that each covers one and only one invalid class until all classes are covered.
- After the AI generated the test cases, I reviewed them to see if the valid and invalid equivalence classes had been covered and if the above rules were followed. If not, I would directly add to them or continue prompting the AI to add any parts I found incomplete. After that, I reviewed all the test cases one more time and selected the most suitable ones, avoiding those that were impossible to execute.
- For subsequent pool requests, I will instruct the AI to perform the same steps as in the first pool.

Pool A: FR-01: Login and account lockout

Test ID	Objective	Input	Test steps	Expected Result	Actual Result
TC1	Verify that a user can log in successfully, receive a JWT token, and access the system when providing valid, matching	Email: <code>test@eshop.com</code> Password: <code>Test1234!</code> Failed Attempts X: 0	Step 1: Open a browser and navigate to the EShop frontend (<code>http://localhost:5173</code>). Step 2: Ensure the backend failed attempts counter for this account is currently less than 3. Step 3: Enter <code>test@eshop.com</code> into	Pass: Login successful, system returns JWT Token.	Login successful, JWT Token was returned: <code>"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6Iiwicm9sZSI6InVzZXliLCJpYXQ891m2n6vE0moqws"</code>

Test ID	Objective	Input	Test steps	Expected Result	Actual Result
	credentials on an active account.		the Email field. Step 4: Enter Test1234! into the Password field. Step 5: Click the login submission button.		
TC2	Verify that the system blocks form submission and prompts the user when the Email field is missing data.	Email: empty Password: Test1234! Failed Attempts X: 0	Step 1: Open a browser and navigate to the EShop frontend (http://localhost:5173). Step 2: Leave the Email field blank. Step 3: Attempt to submit the form. Step 4: Attempt to submit the form.	Fail: UI prevents submission, prompts for Email.	Cannot login and UI prompted for email
TC3	Verify that the system blocks form submission and prompts the user when the Password field is missing data.	Email: test@eshop.com Password: empty Failed Attempts X: 0	Step 1: Navigate to the EShop frontend login screen. Step 2: Enter test@eshop.com into the Email field. Step 3: Leave the Password field blank. Step 4: Attempt to submit the form.	Fail: UI prevents submission, prompts for Password.	Cannot login and UI prompted for password
TC4	Verify that the system completely blocks form submission when no credentials are provided.	Email: empty Password: empty Failed Attempts X: 0	Step 1: Navigate to the EShop frontend login screen. Step 2: Leave both the Email and Password fields blank. Step 3: Attempt to submit the form.	Fail: UI prevents submission, prompts for required fields.	Cannot login and UI prompted for email
TC5	Verify that the HTML5 type="email" validation blocks submission for malformed email strings.	Email: user_at_domain.com Password: Test1234! Failed Attempts X: 0	Step 1: Navigate to the EShop frontend login screen. Step 2: Enter an invalid string (e.g., user_at_domain.com) into the Email field. Step 3: Enter Test1234! into the Password field. Step 4: Attempt to submit the form.	Fail: HTML5 form validation blocks submission.	Cannot login and the UI displays the message "Login failed, please check again".
TC6	Verify that an incorrect password results in a generic failure message and strictly increments the failed attempts counter by 1.	Email: test@eshop.com Password: WrongPassword123! Failed Attempts X: 0	Step 1: Ensure the user account test@eshop.com has a failed attempts counter of 0. Step 2: Navigate to the EShop frontend login screen. Step 3: Enter test@eshop.com into the Email field. Step 4: Enter WrongPassword123! into the Password	Fail: Generic error displayed. Failed attempts counter increments to 1.	Cannot login and the UI displays the message "Login failed, please check again".

Test ID	Objective	Input	Test steps	Expected Result	Actual Result
			field. Step 5: Submit the form.		
TC7	Verify that entering an unregistered email returns the exact same generic error message to prevent account enumeration.	Email: <code>nobody@eshop.com</code> Password: <code>Test1234!</code> Failed Attempts X: 0	Step 1: Navigate to the EShop frontend login screen. Step 2: Enter <code>nobody@eshop.com</code> into the Email field. Step 3: Enter <code>Test1234!</code> into the Password field. Step 4: Submit the form.	Fail: Generic error displayed. Failed attempts counter increments to 1.	Cannot login and the UI displays the message "Login failed, please check again".
TC8	Verify that the system rejects valid logins and enforces a 30-second temporary lockout after 3 consecutive failed attempts.	Email: <code>test@eshop.com</code> Password: <code>Test12345!</code> Failed Attempts X: 3	Step 1: Intentionally fail the login for <code>test@eshop.com</code> 3 consecutive times to push the counter to 3. Step 2: Immediately attempt to log in using the correct credentials (<code>test@eshop.com</code> / <code>Test1234!</code>). Step 3: Submit the form.	Fail: ogin rejected, returns "account temporarily locked for 30s" error.	Cannot login and the UI displays the message "Login failed, please check again".

Pool B: Shopping Cart and Checkout

Test ID	Objective	Precondition	Input	Test Steps	Expected Result	Actual Result	Verdict
TC1	Verify that a valid "percent" type discount code is successfully applied and the final amount is calculated accurately.	User <code>test@eshop.com</code> is logged in with a valid JWT. The user has 0 previous uses of the <code>SAVE10</code> code.	Cart Total: 300,000 đ Coupon Code: <code>SAVE10</code>	1. Navigate to the Checkout page. 2. Verify the cart total is exactly 300,000 đ. 3. Enter <code>SAVE10</code> into the discount code input field. 4. Click the Apply button.	The system accepts the code. A discount of 30,000 đ (10%) is applied. The <code>final_amount</code> is recalculated and displayed as 270,000 đ.	After applying the code <code>SAVE10</code> , the UI displays "Your order has not reached the minimum value of 300,000 VND to apply this code."	FAILED
TC2	Verify that a valid "fixed" type discount code is successfully applied and the final amount is calculated accurately.	User <code>test@eshop.com</code> is logged in with a valid JWT. The user has 0 previous uses of the <code>BIGBUY</code> code.	Cart Total: 500,000 đ Coupon Code: <code>BIGBUY</code>	1. Navigate to the Checkout page. 2. Verify the cart total is 500,000 đ. 3. Enter <code>BIGBUY</code> into the discount code input field. 4. Click the Apply button.	The system accepts the code. A fixed discount of 50,000 đ is applied. The <code>final_amount</code> is recalculated and displayed as 450,000 đ.	After applying the code <code>SAVE10</code> , the UI displays "Your order has not reached the minimum value of 500,000 VND to apply this code."	FAILED
TC3	Verify the system rejects a coupon code that does not exist in the database.	User is logged in.	Cart Total: 500,000 đ Coupon	1. Navigate to the Checkout page. 2. Enter	The system rejects the code and displays an error message	Cannot apply coupon and the UI displayed the following message: "The	PASSED

Test ID	Objective	Precondition	Input	Test Steps	Expected Result	Actual Result	Verdict
			Code: INVALID99	INVALID99 into the discount code input field. 3. Click the Apply button.	indicating the code does not exist.	discount code does not exist or has been invalidated."	
TC4	Verify the system rejects a coupon code if the current date is past its expired_at timestamp.	User is logged in. The user has not used this code before.	Cart Total: 500,000 đ Coupon Code: EXPIRED	1. Navigate to the Checkout page. 2. Enter EXPIRED into the discount code input field. 3. Click the Apply button.	The system rejects the code and displays an error indicating the code has expired (Expired since 2020-01-01).	Cannot apply coupon and the UI displayed the following message: "The discount code has expired."	PASSED
TC5	Verify the system rejects a code if the cart's total value is below the min_order_amount .	User is logged in. The user has not used BIGBUY before.	Cart Total: 490,000 đ Coupon Code: BIGBUY	1. Navigate to the Checkout page. 2. Verify the cart total is 490,000 đ. 3. Enter BIGBUY (requires 500,000 đ). 4. Click the Apply button.	The system rejects the code and displays an error indicating the minimum order amount has not been met.	After applying the code SAVE10 , the UI displays "Your order has not reached the minimum value of 500,000 VND to apply this code."	PASSED
TC6	Verify that guest users (without a valid JWT token) cannot apply discount codes.	The user is browsing as a guest (logged out).	Cart Total: 500,000 đ Coupon Code: BIGBUY	1. Navigate to the Checkout page. 2. Enter BIGBUY into the discount code input field. 3. Click the Apply button.	The system prevents the application of the code and prompts the user to log in.	The system blocks payment before allowing coupon entry and requires login before performing this action.	PASSED
TC7	Verify the system rejects a code if the user has already reached the max_uses_per_user limit.	User is logged in. The database shows the user has already applied the BIGBUY code 1 time (maximum allowed).	Cart Total: 510,000 đ Coupon Code: BIGBUY	1. Navigate to the Checkout page. 2. Enter BIGBUY into the discount code input field. 3. Click the Apply button.	The system rejects the code and displays an error indicating the user has reached the usage limit for this code.	Cannot apply coupon. UI displays "You have used this code once (the limit has been reached)."	PASSED
TC8	Verify that the system accurately calculates the discount_amount using the percentage formula and correctly subtracts it to yield the final_amount .	User is logged in with a valid JWT. The database contains the active code SAVE10 (type: percent, value: 10, min_order: 300,000 đ).	Cart Total: 850,000 đ Coupon Code: SAVE10	1. Navigate to the Checkout page. 2. Verify the cart total is exactly 850,000 đ. 3. Enter SAVE10 into the discount code input field. 4. Click the "Apply" button. 5. Observe the breakdown of	The system calculates the discount amount as: $850,000 * 10 / 100 = 85,000 \text{ đ}$ The system calculates the final amount as: $850,000 - 85,000 = 765,000 \text{ đ}$. The UI clearly displays the Discount (85,000 đ) and the updated Total (765,000 đ).	The code SAVE10 cannot be applied; the total payment remains 850,000 VND.	FAILED

Test ID	Objective	Precondition	Input	Test Steps	Expected Result	Actual Result	Verdict
TC9	Verify that the system accurately applies a fixed flat-rate <code>discount_amount</code> and correctly subtracts it to yield the <code>final_amount</code>	User is logged in with a valid JWT. The database contains the active code <code>VIP100</code> (type: fixed, value: 100,000 đ, min_order: 300,000 đ).	Cart Total: 450,000 đ đ Coupon Code: <code>VIP100</code>	the order totals on the UI.			
				1. Navigate to the Checkout page. 2. Verify the cart total is exactly 450,000 đ. 3. Enter <code>VIP100</code> into the discount code input field. 4. Click the "Apply" button. 5. Observe the breakdown of the order totals on the UI.	The system identifies the discount amount as the fixed value: 100,000 đ. The system calculates the final amount as: 450,000 - 100,000 = 350,000 đ. The UI clearly displays the Discount (100,000 đ) and the updated Total (350,000 đ).	After successfully applying the coupon, the payment amount was reduced to 350,000 VND.	PASSED

Pool C: Web Admin

Test ID	Objective	Precondition	Input	Test Steps	Expected Result	Actual Result	Verdict
TC1	Verify that an Admin can successfully create a new percentage-based discount code when all inputs are valid.	User is logged in as an Admin (<code>role = 'admin'</code>). The code <code>SUMMER20</code> does not exist in the database.	<code>code: SUMMER20</code> <code>type: percent</code> <code>discount_value: 20</code> <code>expired_at: 2026-12-31 23:59:59</code> <code>min_order_amount: 0</code> <code>max_uses_per_user: 1</code>	1. Navigate to the Coupon Management section in the Web Admin portal. 2. Click Add New Coupon . 3. Fill in the form with the input data. 4. Click Save/Create .	The system successfully creates the coupon and displays it in the coupon list. A success notification is shown.	Coupon created successfully, coupon appears in the list of coupons.	PASSED
TC2	Verify that an Admin can successfully create a new fixed-amount discount code when all inputs are valid.	User is logged in as an Admin. The code <code>WELCOMES0K</code> does not exist in the database.	<code>code: WELCOMES0K</code> <code>type: fixed</code> <code>discount_value: 50000</code> <code>expired_at: 2026-12-31 23:59:59</code> <code>min_order_amount: 200000</code> <code>max_uses_per_user: 5</code>	1. Navigate to the Coupon Management section. 2. Click Add New Coupon . 3. Fill in the form with the input data. 4. Click Save .	The system successfully creates the fixed-value coupon and adds it to the list with the correct configuration.	Coupon created successfully, coupon appears in the list of coupons.	PASSED
TC3	Verify the system rejects coupon creation if the Admin enters a code that already exists in the database.	User is logged in as an Admin. The code <code>SAVE10</code> already exists in the system.	<code>code: SAVE10</code> All other fields contain valid values.	1. Navigate to the Coupon Management section. 2. Click Add New Coupon . 3. Enter <code>SAVE10</code> in the code field. 4. Fill all remaining fields with valid data. 5. Click Save .	The system prevents creation and displays a validation error indicating that the coupon code must be unique.	The system did not generate the discount code, and this error prevention measure was already present in the database.	PASSED
TC4	Verify the system enforces the mandatory nature of the coupon code field.	User is logged in as an Admin.	<code>code: (blank)</code> All other fields contain valid values.	1. Navigate to the Add Coupon form. 2. Leave the code field empty. 3. Fill all remaining fields with valid data. 4. Click Save .	Form submission is blocked and a required-field validation error is displayed for the code field.	The system failed to generate the coupon, and the UI is now prompting you to enter the coupon code.	PASSED

Test ID	Objective	Precondition	Input	Test Steps	Expected Result	Actual Result	Verdict
TC5	Verify the system rejects a coupon if the discount value is not a positive number.	User is logged in as an Admin.	<code>discount_value: 0</code> (or <code>-10</code>) All other fields contain valid values.	1. Navigate to the Add Coupon form. 2. Enter valid data for all fields except <code>discount_value</code> . 3. Enter <code>0</code> in the <code>discount_value</code> field. 4. Click Save .	The system rejects the input and displays an error stating that the discount value must be positive.	The system allows admin to create coupons with a value of ≤ 0 .	FAILED
TC6	Verify the system enforces the non-negative boundary for the minimum order amount.	User is logged in as an Admin.	<code>min_order_amount: -50000</code> All other fields contain valid values.	1. Navigate to the Add Coupon form. 2. Enter valid data for all fields except <code>min_order_amount</code> . 3. Enter <code>-50000</code> into the <code>min_order_amount</code> field. 4. Click Save .	The system rejects the input and displays an error stating that the minimum order amount cannot be less than zero.	The system allows creating coupons with a <code>min_order_amount</code> is <code>-50000</code> .	FAILED
TC7	Verify the system rejects a coupon configuration if the maximum uses allowed per user is less than 1.	User is logged in as an Admin.	<code>max_uses_per_user: 0</code> All other fields contain valid values.	1. Navigate to the Add Coupon form. 2. Enter valid data for all fields except <code>max_uses_per_user</code> . 3. Enter <code>0</code> into the <code>max_uses_per_user</code> field. 4. Click Save .	The system rejects the input and displays an error stating that the usage limit must be at least 1.	The system prevents coupon creation, and the UI now requires <code>max_uses_per_user</code> to be greater than or equal to 1.	PASSED
TC8	Verify that the system strictly enforces the allowed enum values for the <code>type</code> field and rejects unsupported values.	User is logged in as an Admin (<code>role = 'admin'</code>). The code <code>WINTER99</code> does not exist in the database.	<code>code: WINTER99</code> <code>type: freeshipping (invalid)</code> <code>discount_value: 20</code> <code>expired_at: 2026-12-31 23:59:59</code> <code>min_order_amount: 0</code> <code>max_uses_per_user: 1</code>	1. Navigate to the Coupon Management section. 2. Click Add New Coupon . 3. Fill in the form with the provided data. 4. If the UI restricts values through a dropdown, intercept the request or use Postman to submit <code>"type": "freeshipping"</code> . 5. Submit the request.	The system rejects the payload and returns a validation error indicating that <code>type</code> must be either <code>percent</code> or <code>fixed</code> . The coupon is not saved to the database.	The system only allows the use of two types of coupons: percent and fixed.	PASSED

POOL D: Mobile App

Test ID	Objective	Input	Test steps	Expected Result	Actual Result	Verdict
TC1	Verify that a user can log in successfully, and access the system when providing valid, matching credentials on an active account.	Email: <code>test@eshop.com</code> Password: <code>Test1234!</code> Failed Attempts X: 0	Step 1: Open a browser and navigate to the EShop frontend (<code>http://localhost:5173</code>). Step 2: Ensure the backend failed attempts counter for this account is currently less than 3. Step 3: Enter <code>test@eshop.com</code> into the Email field. Step 4: Enter <code>Test1234!</code> into the Password field. Step 5: Click the login submission button.	Pass: Login successful, system returns JWT Token.	Login successful	PASSED
TC2	Verify that the system blocks form submission and prompts the user when the Email field is missing data.	Email: empty Password: <code>Test1234!</code> Failed Attempts X: 0	Step 1: Open a browser and navigate to the EShop frontend (<code>http://localhost:5173</code>). Step 2: Leave the Email field blank. Step 3: Attempt to submit the form.	Fail: UI prevents submission, prompts for Email.	Cannot login and UI prompted for email	PASSED

Test ID	Objective	Input	Test steps	Expected Result	Actual Result	Verdict
			Step 4: Attempt to submit the form.			
TC3	Verify that the system blocks form submission and prompts the user when the Password field is missing data.	Email: <code>test@eshop.com</code> Password: empty Failed Attempts X: 0	Step 1: Navigate to the EShop frontend login screen. Step 2: Enter <code>test@eshop.com</code> into the Email field. Step 3: Leave the Password field blank. Step 4: Attempt to submit the form.	Fail: UI prevents submission, prompts for Password.	Cannot login and UI prompted for password	PASSED
TC4	Verify that the system completely blocks form submission when no credentials are provided.	Email: empty Password: empty Failed Attempts X: 0	Step 1: Navigate to the EShop frontend login screen. Step 2: Leave both the Email and Password fields blank. Step 3: Attempt to submit the form.	Fail: UI prevents submission, prompts for required fields.	Cannot login and UI prompted for email	PASSED
TC5	Verify that the HTML5 type="email" validation blocks submission for malformed email strings.	Email: <code>user_at_domain.com</code> Password: <code>Test1234!</code> Failed Attempts X: 0	Step 1: Navigate to the EShop frontend login screen. Step 2: Enter an invalid string (e.g., <code>user_at_domain.com</code>) into the Email field. Step 3: Enter <code>Test1234!</code> into the Password field. Step 4: Attempt to submit the form.	Fail: HTML5 form validation blocks submission.	Cannot login and the UI displays the message "Login failed, please check again".	FAILED
TC6	Verify that an incorrect password results in a generic failure message and strictly increments the failed attempts counter by 1.	Email: <code>test@eshop.com</code> Password: <code>WrongPassword123!</code> Failed Attempts X: 0	Step 1: Ensure the user account <code>test@eshop.com</code> has a failed attempts counter of 0. Step 2: Navigate to the EShop frontend login screen. Step 3: Enter <code>test@eshop.com</code> into the Email field. Step 4: Enter <code>WrongPassword123!</code> into the Password field. Step 5: Submit the form.	Fail: Generic error displayed. Failed attempts counter increments to 1.	Cannot login and the UI displays the message "Login failed, please check again".	PASSED
TC7	Verify that entering an unregistered email returns the exact same generic error message to prevent account enumeration.	Email: <code>nobody@eshop.com</code> Password: <code>Test1234!</code> Failed Attempts X: 0	Step 1: Navigate to the EShop frontend login screen. Step 2: Enter <code>nobody@eshop.com</code> into the Email field. Step 3: Enter <code>Test1234!</code> into the Password field. Step 4: Submit the form.	Fail: Generic error displayed. Failed attempts counter increments to 1.	Cannot login and the UI displays the message "Login failed, please check again".	PASSED
TC8	Verify that the system rejects valid logins and enforces a 30-second temporary lockout after 3 consecutive failed attempts.	Email: <code>test@eshop.com</code> Password: <code>Test12345!</code> Failed Attempts X: 3	Step 1: Intentionally fail the login for <code>test@eshop.com</code> 3 consecutive times to push the counter to 3. Step 2: Immediately attempt to log in using the correct credentials (<code>test@eshop.com</code> / <code>Test1234!</code>). Step 3: Submit the form.	Fail: ;Login rejected, returns "account temporarily locked for 30s" error.	Cannot login and the UI displays the message "Login failed, please check again".	PASSED

BOUNDARY VALUE ANALYSIS

- After performing the Domain Testing step above, I will execute the AI request based on the instructions in the S04_Domain Testing file to perform Boundary Value Analysis, and the test cases must include Test ID, Objective, Precondition (if any), Input, Test steps, and Expected Result.
- After the AI provides its answer, I will review the entire response to check if it is correct and covers all boundary testing cases. If the test cases cannot be executed, I will remove them, or if there are errors, I will make corrections by asking AI which test cases it missed and requesting it generate missing test cases i founded. The system only allows the use of two types of coupons: percent and fixed.

- For subsequent pool requests, I will instruct the AI to perform the same steps as in the first pool.

Pool A: Authentication, Categories, and Products

Test ID	Objective	Precondition	Input	Test Steps	Expected Result	Actual Result
TC_BVA_01	Ensure the system allows a normal login when the failed attempts counter is at its smallest possible value (X = 0).	The account test@eshop.com currently has 0 consecutive failed login attempts.	Email: test@eshop.com Password: Correct Password	1. Navigate to the login page. 2. Enter Email test@eshop.com and the correct Password. 3. Click " Đăng nhập ".	Login is successful, and the system returns a JWT Token.	Login successful, JWT Token was returned: "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6Miwicm9sZSI6Sbbq891m2n6vE0moqws"
TC_BVA_02	Ensure that when the user has failed 2 times (X = 2), the system still allows them to make a 3rd attempt.	The account test@eshop.com currently has 2 consecutive failed login attempts.	Email: test@eshop.com Password: Correct Password	1. Navigate to the login page. 2. Enter Email test@eshop.com and the correct Password. 3. Click " Đăng nhập ".	Login is successful; the system resets the failed attempt counter to 0 and returns a JWT Token.	Cannot login and the UI displays the message "Login failed, ple
TC_BVA_03	Ensure that on exactly the 3rd failed attempt (X = 3), the account gets locked.	The account currently has 2 failed attempts.	Email: test@eshop.com Password: Incorrect Password	1. Navigate to the login page. 2. Enter Email test@eshop.com and an incorrect Password. 3. Click " Đăng nhập ".	Login fails. The system immediately locks the account for 30 seconds and displays an appropriate error message.	Cannot login and the UI displays the message "Login failed, ple
TC_BVA_04	Test just before the unlock threshold to ensure the system does not unlock the account faster than intended (T = 29 seconds).	The account test@eshop.com was just locked due to 3 failed attempts.	Email: test@eshop.com Password: Correct Password Time elapsed: 29 seconds	1. Wait exactly 29 seconds from the logout time. 2. Enter the correct Password. 3. Click " Đăng nhập ".	The system rejects the login and notifies the user that the account is still temporarily locked.	Cannot login and the UI displays the message "Login failed, ple
TC_BVA_05	Ensure the system unlocks exactly at the 30-second boundary as specified (T = 30 seconds).	The account test@eshop.com was just locked.	Email: test@eshop.com Password: Correct Password Time elapsed: 30 seconds	1. Wait exactly 30 seconds from the logout time. 2. Enter the correct Password. 3. Click " Đăng nhập ".	Login is successful; the system unlocks the account, resets the failure counter, and returns a JWT Token.	Cannot login and the UI displays the message "Login failed, ple
TC_BVA_06	Ensure the account	The account test@eshop.com	Email: test@eshop.com	1. Wait 31 seconds from the	Login is successful,	Cannot login and the UI displays the message "Login failed, ple

Test ID	Objective	Precondition	Input	Test Steps	Expected Result	Actual Result
	maintains normal operational status after safely passing the unlock threshold (T = 31 seconds).	was just locked.	Password: Correct Password Time elapsed: 31 seconds	lockout time. 2. Enter the correct Password. 3. Click " Đăng nhập ".	yielding the same outcome as TC_BVA_05.	

Pool B: Shopping Cart and Checkout

Test ID	Objective	Precondition	Input	Test Steps	Expected Result	Actual Result	Verdict
TC_BVA_01	Verify the system rejects the coupon just below the required minimum order threshold.	User is logged in and has not used BIGBUY before.	Cart Total: 499,999 đ Coupon Code: BIGBUY	1. Navigate to the Checkout page. 2. Ensure cart total is exactly 499,999 đ. 3. Enter BIGBUY and click Apply .	The system rejects the code because the condition (Total $\geq$ 500,000) is not met. An error indicating the minimum order amount has not been reached is displayed.	After applying the code SAVE10, the UI displays "Your order has not reached the minimum value of 500,000 VND to apply this code."	PASSED
TC_BVA_02	Verify the system accepts the coupon exactly at the minimum order boundary.	User is logged in and has not used BIGBUY before.	Cart Total: 500,000 đ Coupon Code: BIGBUY	1. Navigate to the Checkout page. 2. Ensure cart total is exactly 500,000 đ. 3. Enter BIGBUY and click Apply .	The system accepts the code, satisfies the $\geq$ condition, and applies the 50,000 đ discount.	After applying the code SAVE10, the UI displays "Your order has not reached the minimum value of 500,000 VND to apply this code."	FAILED
TC_BVA_03	Verify the system safely accepts the coupon just above the minimum order boundary.	User is logged in and has not used BIGBUY before.	Cart Total: 500,001 đ Coupon Code: BIGBUY	1. Navigate to the Checkout page. 2. Ensure cart total is exactly 500,001 đ. 3. Enter BIGBUY and click Apply .	The system accepts the code and applies the 100,000 đ discount.	Coupons was accepted and the UI displayed total payment is 450.001 đ	PASSED
TC_BVA_04	Ensure the system allows application when the user has never used the code before.	User is logged in. The coupon 'VIP100' has not been used twice before.	Cart Total: 300,001 đ Coupon Code: VIP100	1. Enter VIP100 at checkout and click Apply .	Code is accepted because the condition (0 < 2) is true.	TThe system accepts the code and applies the 100,000 đ	PASSED
TC_BVA_05	Ensure the system allows application on the absolute final allowed attempt.	User is logged in. The coupon 'VIP100' has been used once before.	Cart Total: 300,001 đ Coupon Code: VIP100	1. Enter VIP100 at checkout and click Apply .	Code is accepted because the condition (1 < 2) is true. This consumes the final allowed usage.	The system accepts the code and applies the 100,000 đ	PASSED
TC_BVA_06	Catch mis-specified inequality errors and ensure the system blocks the third attempt	User is logged in. The coupon 'VIP100' has been used twice before.	Cart Total: 300,001 đ Coupon	1. Enter VIP100 at checkout and click Apply .	The system rejects the code because the usage limit has been reached (2 < 2 is false).	Cannot apply the coupon, the UI displayed "You have used this code twice (you have reached the limit)."	PASSED

Test ID	Objective	Precondition	Input	Test Steps	Expected Result	Actual Result	Verdict
	after the usage limit is reached.		Code: VIP100				

Pool C: Web Admin

Test ID	Objective	Precondition	Input	Test Steps	Expected Result	Actual Result	Verdict
TC_BVA_01	Verify the system rejects a coupon creation when the discount value is exactly zero, as the requirement strictly states it must be positive.	User is logged in as an Admin. Code TEST01 does not exist.	code: TEST01 discount_value: 0 All other fields valid.	1. Navigate to the Add Coupon form. 2. Enter 0 into the discount_value field. 3. Click Save .	The system rejects the input. A validation error indicates that the discount value must be greater than 0.	The system allows admin to create coupons with a value of 0.	FAILED
TC_BVA_02	Verify the system accepts a coupon creation at the smallest positive integer value.	User is logged in as an Admin. Code TEST02 does not exist.	code: TEST02 discount_value: 1 All other fields valid.	1. Navigate to the Add Coupon form. 2. Enter 1 into the discount_value field. 3. Click Save .	The system successfully creates the coupon with a discount value of 1.	The system allows admin to create coupons with a value of 1.	PASSED
TC_BVA_03	Verify the system rejects a minimum order amount just below the allowed threshold.	User is logged in as an Admin. Code TEST03 does not exist.	code: TEST03 min_order_amount: -1 All other fields valid.	1. Navigate to the Add Coupon form. 2. Enter -1 into the min_order_amount field. 3. Click Save .	The system rejects the input. A validation error indicates that the minimum order amount cannot be less than 0.	The system allows creating coupons with a min_order_amount is -1.	FAILED
TC_BVA_04	Verify the system safely accepts a coupon requiring no minimum order amount.	User is logged in as an Admin. Code TEST04 does not exist.	code: TEST04 min_order_amount: 0 All other fields valid.	1. Navigate to the Add Coupon form. 2. Enter 0 into the min_order_amount field. 3. Click Save .	The system successfully creates the coupon.	The system allows creating coupons with a min_order_amount is 0.	PASSED
TC_BVA_05	Verify the system accepts a minimum order amount just above the boundary.	User is logged in as an Admin. Code TEST05 does not exist.	code: TEST05 min_order_amount: 1 All other fields valid.	1. Navigate to the Add Coupon form. 2. Enter 1 into the min_order_amount field. 3. Click Save .	The system successfully creates the coupon.	The system allows creating coupons with a min_order_amount is 1.	PASSED
TC_BVA_06	Verify the system rejects a usage limit set to zero and catches possible inequality implementation errors.	User is logged in as an Admin. Code TEST06 does not exist.	code: TEST06 max_uses_per_user: 0 All other fields valid.	1. Navigate to the Add Coupon form. 2. Enter 0 into the max_uses_per_user field. 3. Click Save .	The system rejects the input. A validation error indicates that the usage limit must be at least 1.	The system prevents coupon creation, and the UI now requires max_uses_per_user to be greater than or equal to 1.	PASSED
TC_BVA_07	Verify the system accepts a usage limit of exactly 1.	User is logged in as an Admin. Code TEST07 does not exist.	code: TEST07 max_uses_per_user: 1 All other fields valid.	1. Navigate to the Add Coupon form. 2. Enter 1 into the max_uses_per_user field. 3. Click Save .	The system successfully creates the coupon with a single-use restriction.	This system allows admin to create this coupon.	PASSED
TC_BVA_08	Verify the system accepts a usage limit just above the boundary.	User is logged in as an Admin. Code TEST08 does not exist.	code: TEST08 max_uses_per_user: 2 All other fields valid.	1. Navigate to the Add Coupon form. 2. Enter 2 into the max_uses_per_user field. 3. Click Save .	The system successfully creates the coupon.	This system allows admin to create this coupon.	PASSED

Pool D: Mobile App

Test ID	Objective	Precondition	Input	Test Steps	Expected Result	Actual Result	Verdict
TC_BVA_01	Ensure the system allows a normal login when the failed attempts counter is at its smallest possible value (X = 0).	The account <code>test@eshop.com</code> currently has 0 consecutive failed login attempts.	Email: <code>test@eshop.com</code> Password: Correct Password	1. Navigate to the login page. 2. Enter Email <code>test@eshop.com</code> and the correct Password. 3. Click "Đăng nhập".	Login is successful.	Login successful.	PASSED
TC_BVA_02	Ensure that when the user has failed 2 times (X = 2), the system still allows them to make a 3rd attempt.	The account <code>test@eshop.com</code> currently has 2 consecutive failed login attempts.	Email: <code>test@eshop.com</code> Password: Correct Password	1. Navigate to the login page. 2. Enter Email <code>test@eshop.com</code> and the correct Password. 3. Click "Đăng nhập".	Login is successful; the system resets the failed attempt counter to 0.	Cannot login and the UI displays the message "Login failed, please check again".	FAILED
TC_BVA_03	Ensure that on exactly the 3rd failed attempt (X = 3), the account gets locked.	The account currently has 2 failed attempts.	Email: <code>test@eshop.com</code> Password: Incorrect Password	1. Navigate to the login page. 2. Enter Email <code>test@eshop.com</code> and an incorrect Password. 3. Click "Đăng nhập".	Login fails. The system immediately locks the account for 30 seconds and displays an appropriate error message.	Cannot login and the UI displays the message "Login failed, please check again".	PASSED
TC_BVA_04	Test just before the unlock threshold to ensure the system does not unlock the account faster than intended (T = 29 seconds).	The account <code>test@eshop.com</code> was just locked due to 3 failed attempts.	Email: <code>test@eshop.com</code> Password: Correct Password Time elapsed: 29 seconds	1. Wait exactly 29 seconds from the logout time. 2. Enter the correct Password. 3. Click "Đăng nhập".	The system rejects the login and notifies the user that the account is still temporarily locked.	Cannot login and the UI displays the message "Login failed, please check again".	PASSED
TC_BVA_05	Ensure the system unlocks exactly at the 30-second boundary as specified (T = 30 seconds).	The account <code>test@eshop.com</code> was just locked.	Email: <code>test@eshop.com</code> Password: Correct Password Time elapsed: 30 seconds	1. Wait exactly 30 seconds from the logout time. 2. Enter the correct Password. 3. Click "Đăng nhập".	Login is successful; the system unlocks the account, resets the failure counter.	Cannot login and the UI displays the message "Login failed, please check again".	FAILED
TC_BVA_06	Ensure the account maintains normal operational status after safely passing the unlock threshold (T = 31 seconds).	The account <code>test@eshop.com</code> was just locked.	Email: <code>test@eshop.com</code> Password: Correct Password Time elapsed: 31 seconds	1. Wait 31 seconds from the logout time. 2. Enter the correct Password. 3. Click "Đăng nhập".	Login is successful, yielding the same outcome as TC_BVA_05.	Cannot login and the UI displays the message "Login failed, please check again".	FAILED

3. AI GAP ANALYSIS

All test cases were generated by AI. Initially, the AI generated test cases that were sometimes complete, sometimes impractical, and sometimes incomplete. However, after careful review and pointing out these issues, the AI recreated the missing cases according to my requirements.

4. BUG REPORT

4.1 Unable to login after successfully entering the account details for the third time following two consecutive failed login attempts.

Test step:

- 1. Navigate to the login page.
- 2. Enter Email `test@eshop.com` and an correct Password.
- 3. Click "Đăng nhập".

Expected Result: Login is successful; the system resets the failed attempt counter to 0.

Bug: Both web and mobile platforms were unable to log in with the correct email and password after two failed attempts.

4.2 Unable to log in with the correct email and password after 30 seconds and 31 seconds of being locked.

Test step:

- 1. Wait exactly 30 or 31 seconds from the logout time.
- 2. Enter the correct Password.
- 3. Click "Đăng nhập".

Expected Result: Login is successful; the system unlocks the account, resets the failure counter.

Bug: Cannot login and the UI displays the message "Login failed, please check again".

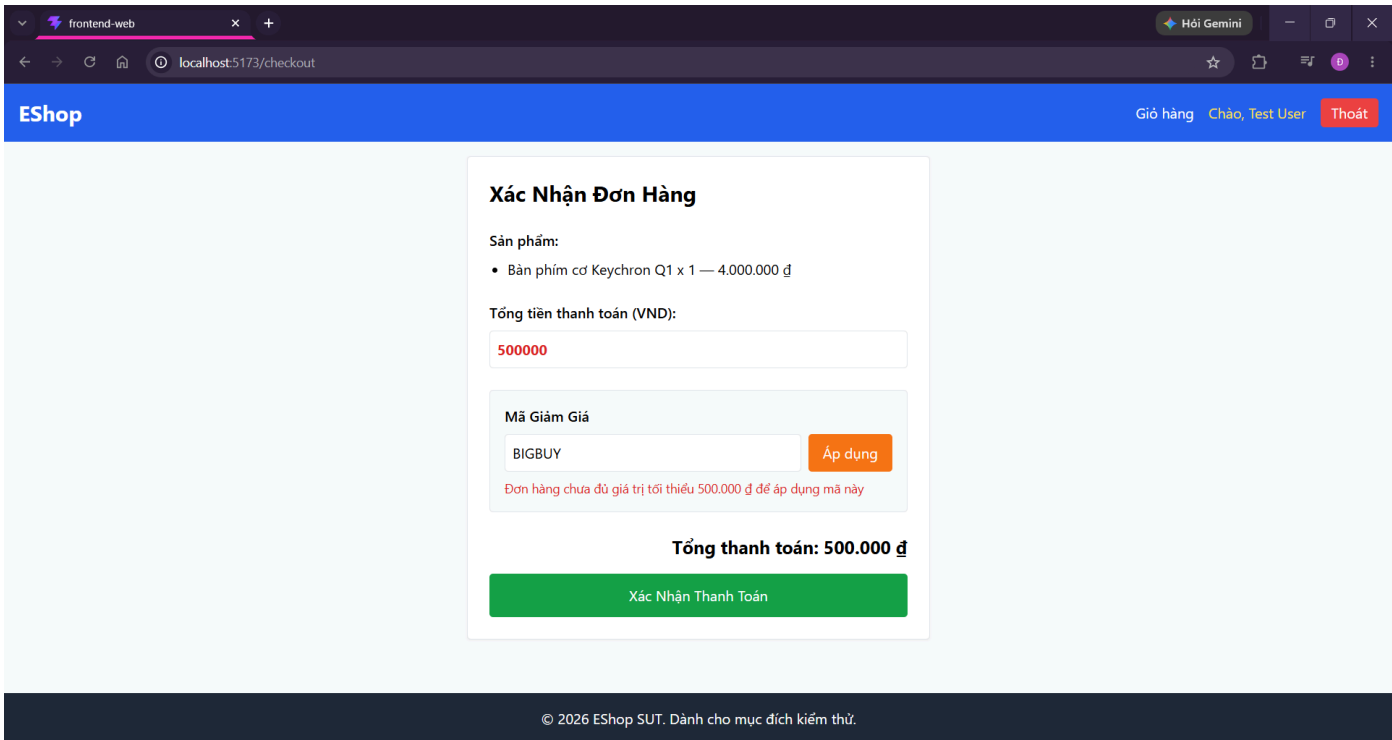
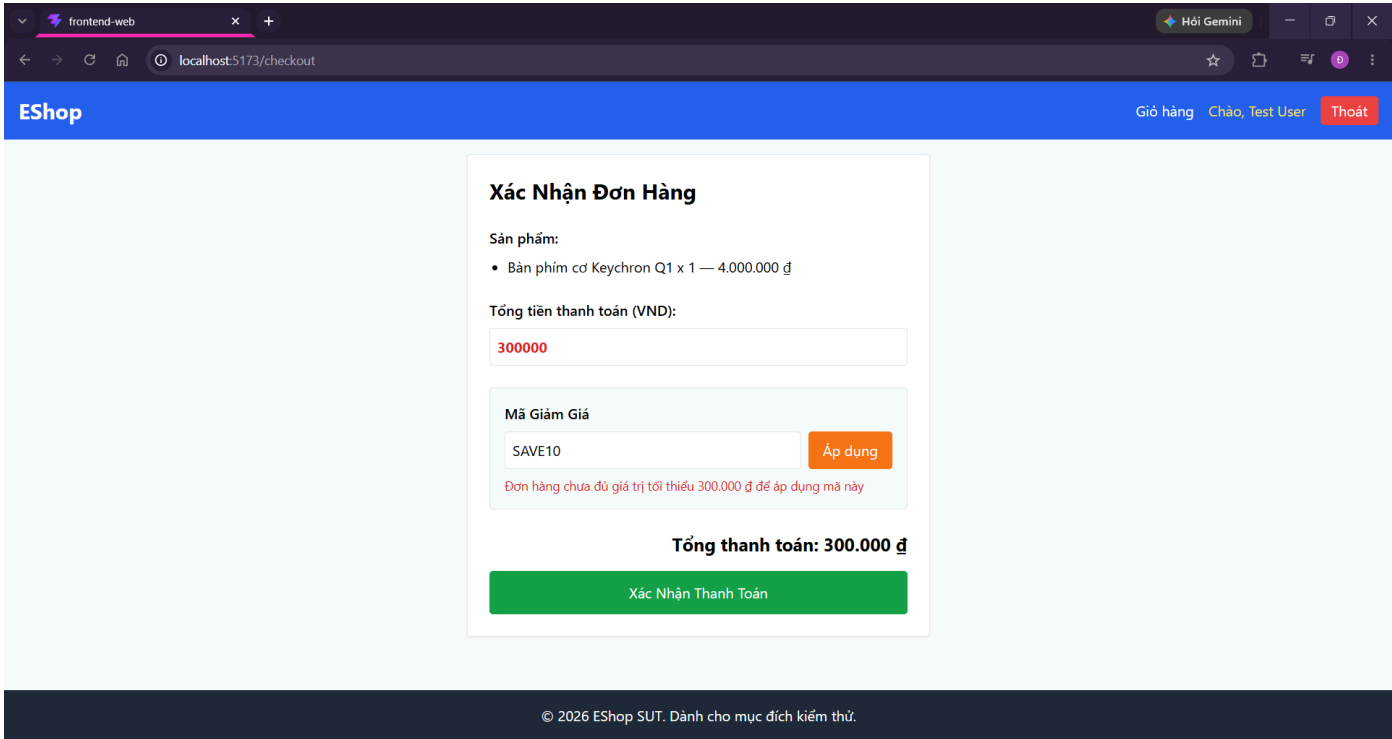
4.3 Coupons cannot be used with a minimum payment amount (equal to the minimum threshold).

Test step: Navigate to the Checkout page.

1. Navigate to the Checkout page.
2. Make sure the total value of your shopping cart exactly equals the minimum threshold of the coupon you wish to use.
3. Enter the coupon with the corresponding minimum payment amount.

Expected Result: The coupon is successfully applied, the total payment amount will be reduced based on the coupon value.

Bug screen:

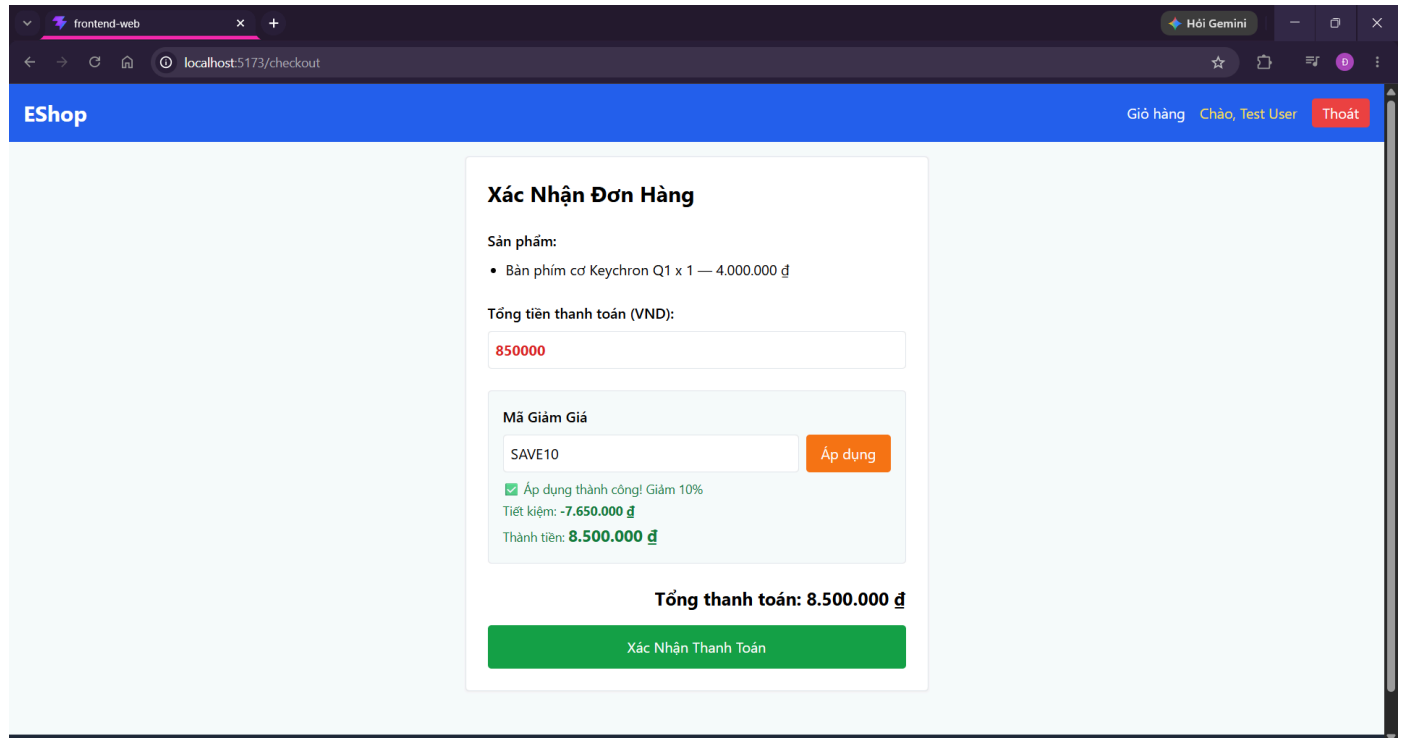


4.4 Percentage coupon codes cannot be used; the discount will not be applied to the total payment.

Test step:

- 1. Navigate to the Checkout page.
- 2. Verify the cart total is 8,500,000 đ.
- 3. Enter SAVE10 into the discount code input field.
- 4. Click the Apply button.

Expected Result: After successfully applying the coupon, the total payment amount will be reduced by 10%.

Bug screen:

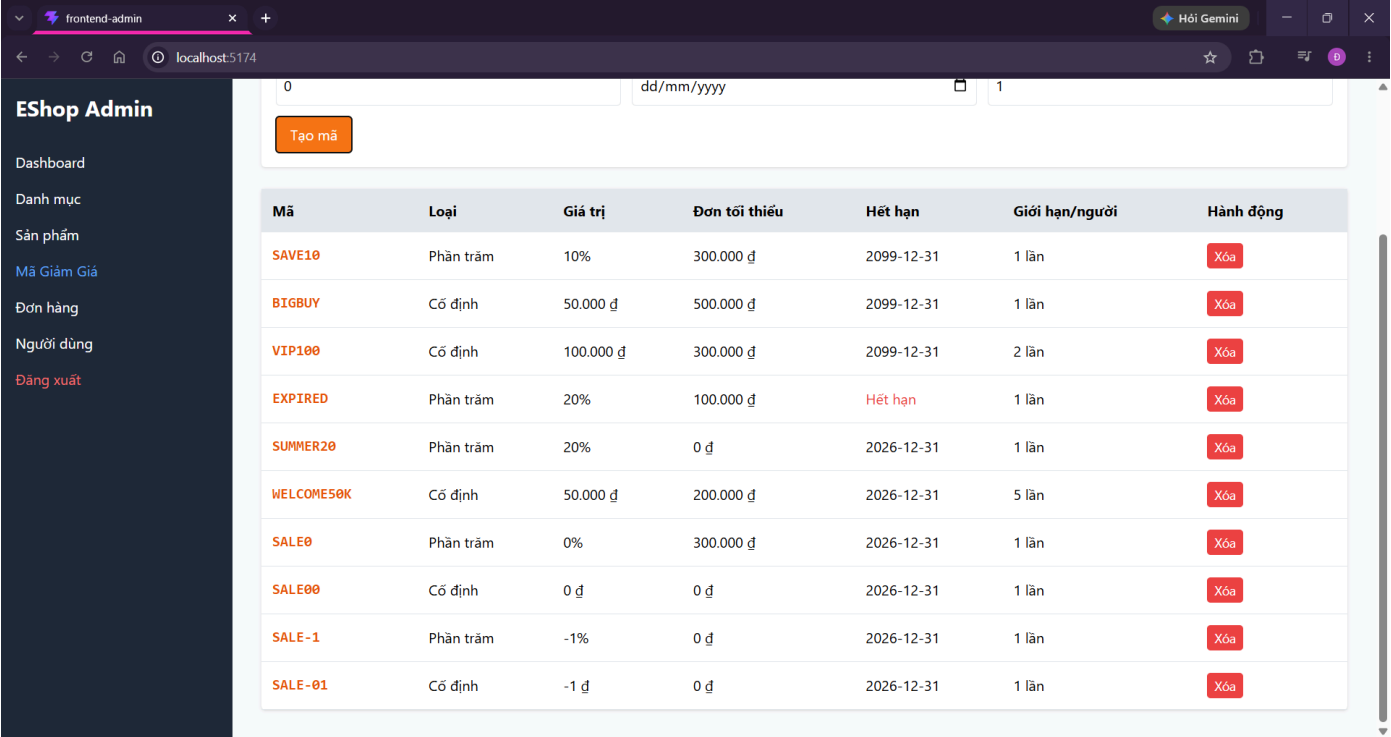
4.5 The system allows creating coupons with a value of ≤ 0 (different from the initial requirements).

Test step:

- 1. Navigate to the Add Coupon form.
- 2. Enter 0 or negative integer into the discount_value field.
- 3. Click Save.

Expected Result: The system rejects the input. A validation error indicates that the discount value must be greater than 0.

Bug screen:



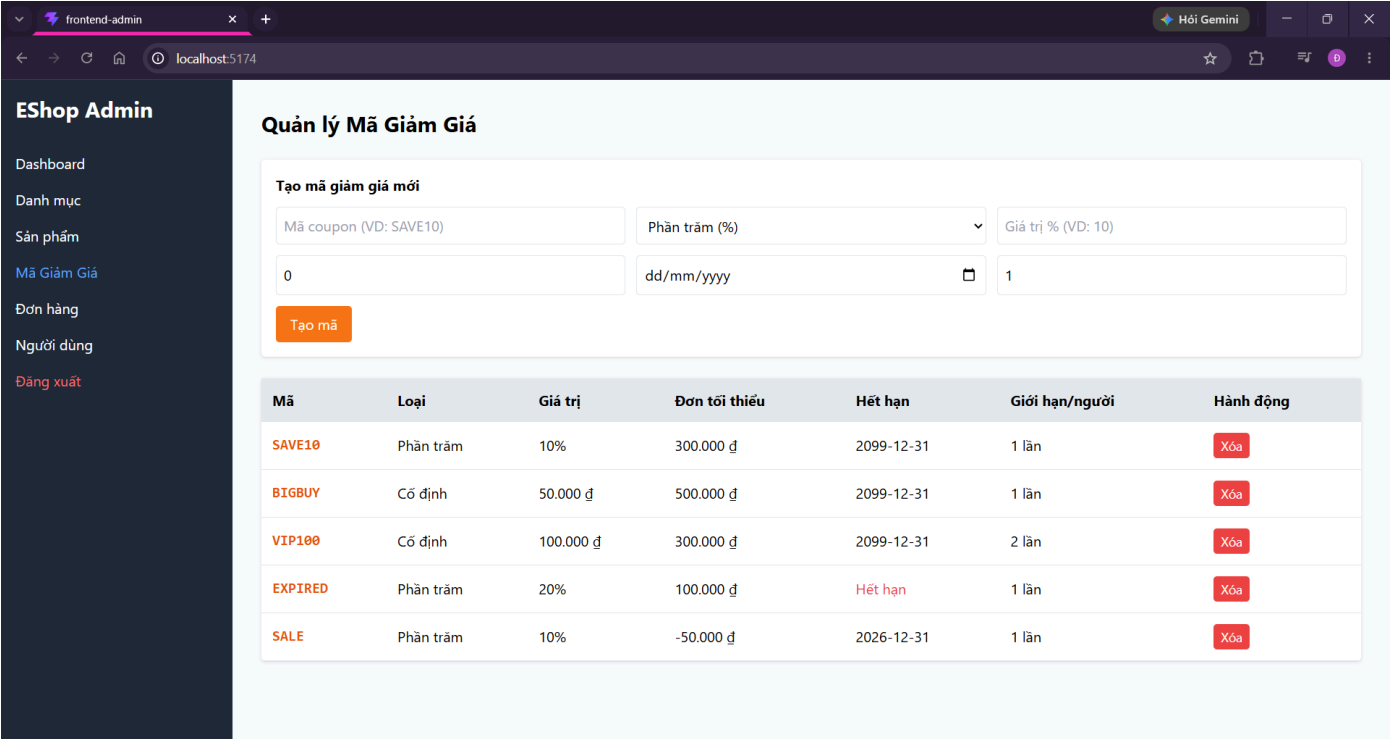
4.6 The system allows creating coupons with a min_order_amount of <0 (different from the initial requirements).

Test step:

- 1. Navigate to the Add Coupon form.
- 2. Enter -50000đ into the min_order_amount field.
- 3. Click Save.

Expected Result: The system rejects the input and displays an error stating that the minimum order amount cannot be less than zero.

Bug screen:



5. AI agent skills

I used AI agent skills by creating the SKILL.md file in Vscode Agent Skills, which can be viewed in the .agents/skills/domain-testing-dat directory.

Demo video: [AI agent skills Demo](#)

6. Appendix

[AI-02 AI Audit Report](#)

[AI Critique](#)

Appendix B

[Self-assessment & Test summary report](#)